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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Bayesian Statistics and Data Analysis

Course information

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Who am I?

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

1. Associate professor in Statistics
2. Teaching: Machine Learning and Bayesian Statistics
3. Research: Machine Learning and Bayesian Statistics
 - 3.1 Bayesian Statistics and computational statistics



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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 - 3.2 Data-centric machine learning



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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 - 3.1 Bayesian Statistics and computational statistics
 - 3.2 Data-centric machine learning
 - 3.3 Statistical inference from textual data



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

1. Associate professor in Statistics
2. Teaching: Machine Learning and Bayesian Statistics
3. Research: Machine Learning and Bayesian Statistics
 - 3.1 Bayesian Statistics and computational statistics
 - 3.2 Data-centric machine learning
 - 3.3 Statistical inference from textual data
 - 3.4 **Applications:** Law, sociology, political science, history, and medicine/biology



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- **Course information**
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Section 1

Course information



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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The aims of this course are that, after this course you should:

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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,



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UNIVERSITET

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Course information

The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,
4. identify and formulate Bayesian probabilistic models for analysis and predictions,



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UNIVERSITET

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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The aims of this course are that, after this course you should:

1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,
4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,



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1. have knowledge in basic concepts, philosophy, and perspectives in Bayesian Statistics,
2. derive posterior distributions in simple situations,
3. derive and use predictive distributions,
4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,
7. have knowledge about and be able to use Bayesian methods for model comparisons,

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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UNIVERSITET

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4. identify and formulate Bayesian probabilistic models for analysis and predictions,
5. estimate models using contemporary (state-of-the-art) computer-based methods for posterior approximations,
6. understand and use basic principles for decisions under uncertainty,
7. have knowledge about and be able to use Bayesian methods for model comparisons,
8. be able to critically evaluate Bayesian methods,

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



UPPSALA
UNIVERSITET

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7. have knowledge about and be able to use Bayesian methods for model comparisons,
8. be able to critically evaluate Bayesian methods,
9. report, orally and in writing, a Bayesian statistical analysis

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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Pre-requisites

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

- Basic probability theory
 - probability, probability density, distribution
 - sum, product rule, and Bayes' rule
 - expectation, mean, variance, median



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

- Basic probability theory
 - probability, probability density, distribution
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 - expectation, mean, variance, median
- Basic linear algebra and calculus



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UNIVERSITET

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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

- Basic probability theory
 - probability, probability density, distribution
 - sum, product rule, and Bayes' rule
 - expectation, mean, variance, median
- Basic linear algebra and calculus
- Basic visualisation techniques (R or Python)
 - histogram, density plot, scatter plot
- *Note!* This is a masters course in Statistics.



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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First assignment is a recap.



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Who are you? (very brief)

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

1. Name
2. What is your expectation on this course?
3. Why did you choose it? (if you chose it)
4. What do you hope to get out of it?



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Course Outline

- **Course information**
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Main components:

- Core Content (8-9 lecture blocks)



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Course Outline

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)



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Course Outline

Main components:

- Course information
 - Assignments
 - Mini-project
 - Practicalities
 - Examination
 - Course improvements
- Core Content (8-9 lecture blocks)
 - Assignments (8 individual assignments)
 - Oral exams connected to the assignments



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UNIVERSITET

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- Course information
- Assignments
- Mini-project
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- Examination
- Course improvements

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)
- Oral exams connected to the assignments
- Mini-project: own Bayesian data analysis or **paper replication track** (1-3 students)



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- Course information
- Assignments
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- Course improvements

Main components:

- Core Content (8-9 lecture blocks)
- Assignments (8 individual assignments)
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Exact dates and details; see the course page.



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Core Content

- Every week: lectures (approx. 2-4h)
 - Online video material and reading assignments (approx. 2-4h, 50-90 pages a week)
 - Lecture(s): present overall theory and content (overview, focus on learning concepts)
 - Assignment(s): Computational and theoretical individual work. (approx. 12-16h). **Deadline Sundays 23.59.**
Start monday morning every week!



- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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Start monday morning every week!
- Recommended workflow for each week
 - Do the reading assignments
 - Watch the videos (optional)
 - Do self-study exercises
 - Start with the assignment
 - Attend lecture (bring questions!)
 - Attend Zoom datalabs (bring questions! Helps with debugging Stan code)
 - Submit the assignment
 - Review the oral exam questions and practice explaining answers in groups



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UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Section 2

Assignments



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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Assignments

1. Core components and concepts and state-of-the-art methods



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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Assignments

1. Core components and concepts and state-of-the-art methods
2. *Warning!* There might be bugs in the assignments!



- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Assignments

1. Core components and concepts and state-of-the-art methods
2. *Warning!* There might be bugs in the assignments!
3. All labs can be turned in a three times (See Studium for details), but only first time can give VG points:
 - 3.1 The week of the assignment
 - 3.2 The last day of the course
 - 3.3 2-4 weeks after the course



- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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LLM use and assignment integrity

- **New this year!** You may use LLMs like fellow students: to learn, ask for explanations, and get feedback.

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements



UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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UNIVERSITET

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- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

- **New this year!** You may use LLMs like fellow students: to learn, ask for explanations, and get feedback.
- LLMs may not produce submitted solutions.
- You must be able to explain every part of your submitted work orally.
- LLMs are not allowed at all for the oral exam questions.
- Assignment PDFs may contain hidden machine-readable integrity text. Do not copy or upload the full assignment text into any LLM.
- If we suspect that the assignment has been done using an LLM, it will be failed.



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Oral exams

- Three oral exams during the course:
 1. Assignments 1–3
 2. Assignments 4–6
 3. Assignments 7–8

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements



UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Oral exams

- Three oral exams during the course:
 1. Assignments 1–3
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 3. Assignments 7–8
- For each covered assignment: one random question from the oral exam document.



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UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Three oral exams during the course:
 1. Assignments 1–3
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- For each covered assignment: one random question from the oral exam document.
- If the first answer is not sufficient, a second question from the same assignment is drawn.



UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- For each covered assignment: one random question from the oral exam document.
- If the first answer is not sufficient, a second question from the same assignment is drawn.
- Passing the oral exam is required to receive VG for the corresponding assignment.



- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Oral exams

- Three oral exams during the course:
 1. Assignments 1–3
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 3. Assignments 7–8
- For each covered assignment: one random question from the oral exam document.
- If the first answer is not sufficient, a second question from the same assignment is drawn.
- Passing the oral exam is required to receive VG for the corresponding assignment.
- Each oral exam has three attempts in total: one ordinary opportunity and two re-examination opportunities.



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UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

R vs Python and Colab

- We strongly recommend using R in the course as there are more packages for Stan and statistical analysis in general in R



UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

R vs Python and Colab

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- If you are already fluent in Python, but not in R, then using Python may be easier, but it can still be useful to learn R



UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- All data for the assignments are included in the R package. You can access the data data or data-raw in the [R package folder](#) in the repo if you need to get the data without using the bsda R package.



UPPSALA
UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

R vs Python and Colab

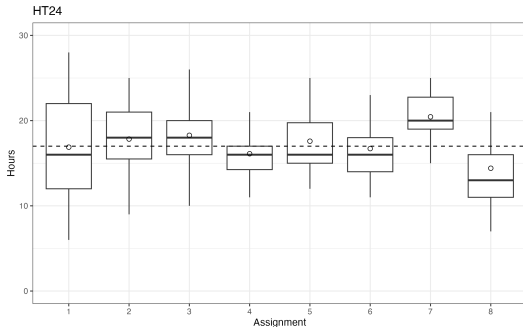
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- We supply a [Google Colab template](#) with everything pre-installed.
- We also supply a knitr $\text{\LaTeX}$ [template](#) suitable for [Overleaf](#).



- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Course Workload

- Conclusions from 2024
 - Slightly too large workload in assignment 7



Student course workload



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UNIVERSITET

- Course information
- **Assignments**
- Mini-project
- Practicalities
- Examination
- Course improvements

Stan

- Stan is a probabilistic programming framework (PPF) and ecosystem
- 40+ developers, 100+ contributors, 100K+ users
- R, Python, Julia, Scala, Stata, Matlab, command line interfaces
- More than 200 R packages using Stan
- Many packages to support diagnostics and workflow
- Can be used for frequentist inference as well
- Alternative PPF exists, Turing (Julia), Pyro (PyTorch), etc.



mc-stan.org



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- Course information
- Assignments
- **Mini-project**
- Practicalities
- Examination
- Course improvements

Section 3

Mini-project



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UNIVERSITET

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.



UPPSALA
UNIVERSITET

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Supply a one-page project proposal in the middle of the course.



UPPSALA
UNIVERSITET

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- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.
- Supply a one-page project proposal in the middle of the course.
- Ideally, use a model not presented in this course.
- Project will last two weeks (half time) - but start earlier.
- The project should use Stan.
- Approximate 40 hours of work *per student*.



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UNIVERSITET

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Supply a one-page project proposal in the middle of the course.
- Ideally, use a model not presented in this course.
- Project will last two weeks (half time) - but start earlier.
- The project should use Stan.
- Approximate 40 hours of work *per student*.
- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)



- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
- 2-3 students.
- Supply a one-page project proposal in the middle of the course.
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- Project will last two weeks (half time) - but start earlier.
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- The first author is corresponding author



- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Mini-project

- See project instructions on webpage for details.
- Two possible tracks: ordinary data analysis on real data, or paper replication.
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- Approximate 40 hours of work *per student*.
- The project should result in a 4 page report (PDF) using the ICML LaTeX template (see course page).
- Project oral presentation (10-15 minutes)
- The first author is corresponding author
- *New this year:*
 - Paper replication track: open paper + Stan implementation + reproduce main result + prepare for posteriordb.
 - Mini-project discussions: after lectures and during lecture breaks.



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- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

Section 4

Practicalities



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Practicalities

- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

- Course page/content: Github – please do a PR or submit an issue if something is wrong!

<https://github.com/MansMeg/BSDA/issues>



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Practicalities

- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

- Course page/content: Github – please do a PR or submit an issue if something is wrong!
<https://github.com/MansMeg/BSDA/issues>
- Communication: Studium



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Practicalities

- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

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<https://github.com/MansMeg/BSDA/issues>
- Communication: Studium
- Schedule: Time Edit/Studium
- Assignments submissions: Studium



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Practicalities

- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

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- Schedule: Time Edit/Studium
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- Acknowledgements: Aki Vehtari



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Practicalities

- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

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- Communication: Studium
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- Assignments submissions: Studium
- Acknowledgements: Aki Vehtari
- Teaching assistant: Väinö Yrjänäinen

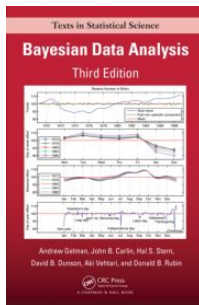


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- Course information
- Assignments
- Mini-project
- **Practicalities**
- Examination
- Course improvements

Literature

- Book: Gelman, Carlin, Stern, Dunson, Vehtari & Rubin: Bayesian Data Analysis, Third Edition. (online pdf available)



- Additional articles, youtube videos and blog posts (see reading list per week)



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- Course information
- Assignments
- Mini-project
- Practicalities
- **Examination**
- Course improvements

Section 5

Examination



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Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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- Course information
- Assignments
- Mini-project
- Practicalities
- **Examination**
- Course improvements

Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed
2. To pass with distinction (VG): 7/10 VG points



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- Course information
- Assignments
- Mini-project
- Practicalities
- **Examination**
- Course improvements

Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed
2. To pass with distinction (VG): 7/10 VG points
3. If everything is correct in an assignment ($\geq 90\%$), 1 VG point is awarded *on the first submission deadline*.



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- Course information
- Assignments
- Mini-project
- Practicalities
- **Examination**
- Course improvements

Examination

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4. Assignment VG points require passing the oral exam for that assignment.



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- Course information
- Assignments
- Mini-project
- Practicalities
- **Examination**
- Course improvements

Examination

1. To pass (G): All labs, oral exams, mini-project, and project review need to be passed
2. To pass with distinction (VG): 7/10 VG points
3. If everything is correct in an assignment ($\geq 90\%$), 1 VG point is awarded *on the first submission deadline*.
4. Assignment VG points require passing the oral exam for that assignment.
5. The mini-project is worth 2 VG-points (if it is passed with distinction).
6. Ph.D. students: I suggest you get VG to pass the course. Make the project a potential paper.
7. Reassessment of grades (supply form to course admin)
8. Failing the course: You will need to redo all assignments and mini-project next year.
9. LLMs may be used for learning, explanations, and feedback, but **not to produce submitted solutions**.



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- **Course improvements**

Section 6

Course improvements



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Course improvements since last time

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

- Oral exams
- New idea (test) of mini project.



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- **Course improvements**

Section 6

Course improvements



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Comments from previous students

- Put in the work, it is worth it.

- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

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- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.
- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect



- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Comments from previous students

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- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.
- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect
- Start the assignments as soon as possible. It is perfectly reasonable to be done with the assignments before the weekend, so aim for that.



- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- Course improvements

Comments from previous students

- Put in the work, it is worth it.
- There is alot to do but it will be worth it if you really put in work.
- Be prepared that the tempo and workload is very high.
- Attend the lectures and start with the assignments early. Because they are really fun if you get passed the time pressure aspect
- Start the assignments as soon as possible. It is perfectly reasonable to be done with the assignments before the weekend, so aim for that.
- Do the course properly, it is worth the effort



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- Course information
- Assignments
- Mini-project
- Practicalities
- Examination
- **Course improvements**

Questions?

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